

SHERAZ HASSAN

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EDUCATION

PhD in Computer Science

Georgia Institute of Technology

Aug 2023 – Present

Atlanta, GA, USA

Relevant Coursework: Machine Learning, Computer Vision, Deep Learning, Theoretical Machine Learning, Ubiquitous Computing, Health Sensing and Intervention

Bachelors in Electrical Engineering

Lahore University of Management Sciences

Sep 2017 – May 2021

Lahore, Pakistan

Relevant Coursework: Deep Learning, Applied Probability, Principles and Techniques of Data Science, Embedded Systems, Signal and Systems

WORK EXPERIENCE

Graduate Research Assistant

Georgia Institute of Technology

Aug 2023 – Present

Atlanta, GA, USA

- Utilizing resource-efficient machine learning and deep learning solutions for biomedical signals and low-cost hardware at the edge.
- Designing low-cost wearable computing hardware, showcasing expertise in sensor integration, system design, and programming microcontrollers for health monitoring and interventions.

Research Assistant

Smart Data Systems and Applications Laboratory (SDSA)

Jun 2021 – Jul 2023

Lahore, Pakistan

- Engineered an affordable healthcare device and a real-time human activity sensing system, while also implementing deep learning models for data analysis and post-processing.
- Led cross-functional teams in designing ML models for diverse applications, including taxi demand prediction, robust heart rate estimation, noise cancellation, and environmental sound classification.

Co-Researcher

Students as Co-Researchers (ScR) Program

Jul 2020 – Sep 2020

Lahore, Pakistan

- Explored healthcare applications of Photoplethysmography (PPG), emphasizing its potential in wearable health technologies.
- Improved heart rate estimation accuracy by mitigating motion artifacts in PPG signals.

TEACHING EXPERIENCE

Georgia Institute of Technology

Artificial Intelligence | Graduate Teaching Assistant

Aug 2025 – Present

Continuing as a TA for the Graduate AI course, supporting instruction, grading, and technical mentoring.

Georgia Institute of Technology

Health Sensing and Interventions | Graduate Teaching Assistant

May 2025 – Aug 2025

Assisted in a hands-on, project-based course exploring digital health technologies and physiological data.

Georgia Institute of Technology

Artificial Intelligence | Graduate Teaching Assistant

Jan 2024 – May 2025

Served as a Teaching Assistant for the Graduate Artificial Intelligence course over multiple semesters, with instruction led by Dr. Thad Starner, Dr. Thomas Ploetz, and Dr. Rodrigo in different terms. Supported over 300 students through office hours, grading, and technical guidance on assignments.

Lahore University of Management Sciences

Introductory Engineering Laboratory | Teaching Assistant

Sep 2022 – Dec 2022

Conducted lab sessions as a Teaching Assistant for over 60 students, offering support in instrumentation, embedded systems, and foundational engineering projects.

PROJECTS

Puffpacket | *C, nRFgo Studio, Keil IDE, Eagle, Zephyr, Bluetooth Low Energy, Low-Cost Design, I2C, SPI*

- Co-designed a low-cost BLE-enabled device for e-cigarettes to detect puffs for behavioral interventions.
- Wrote complete firmware in Zephyr for event sensing, Bluetooth, and low-power sleep state management.

STEF-DHNET | *Python, Tensorflow, Pytorch, Folium, CNN, LSTM, Time Series Data*

- Developed a grid-based deep learning model, merging Long Short Term Memory (LSTM) and Convolutional Neural Networks (CNN), to forecast ride-hailing service demand and supply, incorporating external variables.
- Deployed the model for a multinational ride-hailing platform, while the research paper is presently under review.

Does the plate look correct? | *Python, CV, YOLOv8*

- Retrained the state-of-the-art YOLOv8 for precise segmentation and classification of food items on a plate.
- Developed an end-to-end app for a restaurant QA table, matching plates with menu items for a similarity score.

TinyML Benchmarking for Edge Deployment | *Python, TensorFlow, TFLite Micro, CNN, MobileNet, QAT, PTQ*

- Designed a standardized benchmarking for TinyML using classification and detection tasks on microcontrollers.
- Optimized lightweight models via quantization techniques, achieving significant size reduction with minimal accuracy tradeoff.

EdgeGuard: On-Device Entry System | *Python, TensorFlow, Multitenant Models, Knowledge Distillation*

- Designed a smart system on Nano BLE that detects people via image classification and responds to commands.
- Deployed models for real-time image and audio inference, using knowledge distillation to compress the image classifier for high accuracy with minimal memory footprint.

RKF and Extreme Learning Machines | *MATLAB, Python, Kalman Filters, Extreme Learning Machines*

- Presented a novel approach for heart rate estimation from PPG signals for real-time processing in wearable devices.
- Achieved the lowest computational cost among state-of-the-art algorithms.

Sound Classification | *Python, MATLAB, TensorFlow, tkinter, Simulink, log mel spectrogram, CNN*

- Implemented a Deep CNN model that uses log mel spectrogram for environmental sound classification.
- Deployed a user-friendly graphical user interface for sound input and real-time cancellation after classification.

Robustness of Heart Rate Estimation | *Python, Pytorch, Adversarial Attacks, Regularization*

- Assessed the vulnerability of state-of-the-art heart rate estimation algorithms, like CorNet, to adversarial attacks.
- Implemented a curvature regularization solution enhancing model robustness against ML attacks such as FGSM and CW.

SKILLS

Programming Languages: Python, C/C++, MATLAB, R, SQL

Frameworks & Libraries: TensorFlow, PyTorch, Keras, Scikit-learn, NumPy, SciPy, Pandas, OpenCV, Matplotlib, Streamlit, Flutter, BeautifulSoup, Folium

Development Tools & Platforms: Git, VS Code, Jupyter Notebook, Google Colab, Arduino IDE, STM32CubeIDE, Keil MicroVision, MiKroC, PIC C, nRFgo Studio, Simplicity Studio, QT Designer, Eagle, Autodesk Fusion 360

PUBLICATIONS

“SkinSpex: A Portable Speckle Imaging Prototype for Multiple Skin Biomarker Detection” **Sheraz Hassan**, Kefan Song, Alexander T Adams. IEEE 21st International Conference on Body Sensor Networks (BSN), 2025.

“PuffEM: An E-cigarette Sleeve for Estimating User Nicotine Intake.” Yiyang Wang, Rishabh Goel, **Sheraz Hassan**, Taegen J Doscher, Shilin Wang, Lexington Allen Whalen, Aditya S Gandhi, Yaman S Sangar, Alex Cabral, Xuhai Xu, Josiah Hester, Alexander T Adams. ACM/IEEE Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE), 2025.

“STEF-DHNet: Spatiotemporal External Factors Based Deep Hybrid Network for Enhanced Long-Term Taxi Demand Prediction.” **Sheraz Hassan**, Muhammad Tahir, Momin Uppal, Zubair Khalid, Ivan Gorban, Selim Turki. *arXiv preprint arXiv:2306.14476*, 2023.

ACADEMIC SERVICE

- Reviewer for **CHI 2024** – ACM Conference on Human Factors in Computing Systems. Participated in both initial and revise-and-resubmit rounds, providing detailed, constructive feedback.
- Reviewer for **IEEE Body Sensor Networks (BSN) 2024** – Evaluated research in biomedical systems and haptics with a focus on health sensing and wearable technologies.